INNOVATIVE PRACTICE

The contribution of simulations to the practical work of Foundation Physics students at the University of Limpopo

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Abstract

Purpose – Practical work is regarded as an essential part of learning; hence most tertiary institutions have included a practical component in their physics courses. There is a concern about the effectiveness of the practical work in most universities. The purpose of this study is to assess the contributions of simulations on 20 Foundation Physics students' practical work at the University of Limpopo.

Design/methodology/approach – Two tests, "Determining and interpreting resistive electric circuits concepts test" and the "Test of integrated science process skills" were used. A class test, observation sheet and worksheets for students' practical work were analyzed. Interviews with a selected group of students were also conducted.

Findings – Results indicated that the simulations contributed positively on students' understanding of electric circuits. However, the study revealed that the students who did simulations do not differ from those who did not do the simulations with regards to the development of process skills.

Originality/value – The value of this research is the application of simulation in physics with specific testing methods. In addition, the research is unique in a South African context.

Keywords Simulation, Students, Universities, South Africa

Paper type Research paper

Introduction

There is an educational gap that exists between high schools and tertiary institutions, especially the universities in South Africa (Grayson, 1996). Grayson argues that the gap is wider for students coming from black schools. Through support programmes (e.g. Foundation), the institutions can help the students who were not able to meet the entry requirements because of their poor performance (The International Panel, 2004 cited in Mundalamo, 2006). The University of Limpopo is one of the universities that offer a foundation programme. Foundation Physics is one of the courses offered and students are expected to do practical work as part of their learning. The use of laboratories as part of science teaching and learning is central and plays an important

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role in the education of future scientists and in particular physicists (Corter *et al.*, 2007). Several years ago Arons (1993, p. 278) cited in Trumper (2003) wrote:

> The usefulness and effectiveness of the introductory laboratory have been bones of contention in physics teaching as far as one cares to go back in the literature. Laboratory instruction is costly, and, since its effectiveness has been difficult to substantiate compellingly, some responsible administrators have viewed it is a luxury we cannot afford. Yet most physicists have a deeply rooted, intuitive feeling that laboratory experience is essential to learning and understanding our subject, and they fight hard for maintaining such instruction in the face of frequently expressed doubts and occasionally formidable opposition.

It is hard to imagine learning to do science, or learning about science, without doing laboratory or fieldwork. Laboratories are wonderful venues for teaching and learning science. They offer students opportunities to think about, discuss and solve real problems (Trumper, 2003). Over the years, many scientists have argued that science cannot be meaningful to students without worthwhile practical experiences in the laboratory, that is the reason why all physics I courses in South Africa have a practical component. A productive laboratory is considered to be a learner-centered environment, one that promote students' curiosity, encourages students to ask questions, develops creativity and promotes understanding (Aladejana and Aderibigbe, 2007). The laboratory setting is one major avenue for the learner to actively carry out experiments and construct new information onto his/her existing mental framework for meaningful learning to occur (Sherman, 1995 cited in Aladejana and Aderibigbe, 2007). The question that was raised is whether the laboratory at the University of Limpopo is as good as the one described by Aladejana and Aderibigbe. Therefore, the contribution of simulations to the practical work done by Foundation Physics students at the University of Limpopo was investigated. This research seeks to answer the following question:

What is the contribution of simulations to the practical work done by Foundation Physics students at the University of Limpopo?

The following are the sub-questions:

(1) In what way could simulations contribute to students' understanding of electric circuits?

(2) In what way could simulations contribute to the development of science process skills?

This study was done with 20 Foundation Physics students from Medunsa Campus of the University of Limpopo (UL). UL was established on the 1 January 2005 after the merger of the then University of the North (UN) and the Medical University of South Africa (MEDUNSA). The mission statement of the Physics Department at Medunsa Campus of the University of Limpopo states that: "We develop human capacity to meet the needs of industry, public sector and community". In order to meet the needs of industry, science students have to go through laboratory practice. Students at the Medunsa Campus are expected to complete both a practical and theoretical component successfully to be credited Foundation Physics at the end of the year. The practical work is conducted as a three hours session once a week. Students are provided with a practical manual that outlines what is expected from them, like the laboratory rules and all the experiments that are going to be performed throughout the semester.

The researcher has been observing what was happening in the physics laboratories since 2004. The number of students entering the physics stream has been increasing every year because the university decided to lower the admission requirements since the merger of the two universities. This makes learning difficult in the laboratory because of the limited number of equipments available. Owing to this constraint, students end up working in groups of four to five students per experiment. When students are more than three in a group, some become inactive, as a result very little learning occurred; and then the idea of introducing simulations in the laboratory originated.

The reason for choosing electric circuits is because students experience problems with electric circuits. For example, McDermott (1993) did a study where 500 first year university students were given three simple circuits, one circuit had a single bulb; another had two bulbs in series; the third had two bulbs in parallel. Students were asked to rank the five bulbs according to relative brightness and to explain their reasoning. This comparison required no calculations. Only 15 per cent of the students gave the correct ranking (McDermott, 1993).

Theoretical framework

Constructivism is the accepted paradigm used in education today (Lemmer *et al.*, 2008). A key aspect to this theory is that knowledge is constructed not transmitted and it is the theory of learning not of teaching. Learning may occur through the addition of knowledge to current concepts, creation of new concepts or major modification of current concepts (Bybee, 2002). Through the process of meaningful learning (Ausubel, 1963) instruction should build on learners' prior knowledge. Ausubel (1963) noted a distinction between two types of learning, namely: receptive and discovery. Classroom learning has been traditionally dominated by receptive/rote learning, whereby what is to be learned is presented to, rather than discovered by the learner. Laboratory work can be regarded as a receptive/meaningful learning whereby the learner is presented with concept that is to be learned through simple practical activities.

Methods and data collection

Study sample

A total of 52 students were enrolled for Foundation Physics. These students were divided into two groups during the practical sessions, one group (24 students) was doing electric circuits and the other group (28 students) was doing magnetism. Out of 24 students, 23 volunteered to take part in the research, but three students did not come for all the tests, hence 20 students participated in the research. These students' were arranged in alphabetically according to their surnames and the first one was taken in the experimental group and the second one to the control group; at the end ten students were in the experimental group and the other ten students were in the control group.

Instruments

Five instruments listed below were used for data collection.

Determining and interpreting resistive electric circuit concepts test (DIRECT). The DIRECT was developed to "evaluate students' understanding of a variety of direct current resistive electric circuits concepts". The DIRECT test has been designed for use with high school and college/university students (Engelhardt and Beichner, 2004). The test consists of 29 multiple choice questions, but in this study, the researcher included an option where students were expected to explain their choices in order to prevent students from guessing the correct answer but to motivate their option as well as to evaluate their level of understanding.

Test of integrated science process skills (TISP). In this study, TISP developed by Kazeni (2005) was used and the reason for using TISP is to test students' integrated science process skills. Since TISP was developed in South Africa, it does not use foreign examples and technical terms. The students are expected to develop some skills in the laboratory when performing the experiments.

Worksheets. Students have to take an Electricity and Magnetism module during the second semester of their academic year. The researcher developed the worksheets for the practical work of this module. The same worksheets were used for both simulations and laboratory work with the intention of assessing the contribution of simulations on the experimental group. The questions on the worksheets were testing students' conceptual understanding rather than calculations.

Observation sheet. During the practical session in the laboratory as students were performing the experiments, the researcher made some observation to check whether there was a difference between the two groups in terms of the development of science process skills. The basic process skills were also checked but this study was focusing mainly on the five namely; identifying and controlling variables, stating hypotheses, operational definitions, graphing and interpreting data and experimental design. An observation sheet was drafted to include both experimental and control group.

Class test. Students did electricity and magnetism as part of their theory work. The researcher was not part of theory teaching, but requested the lecturer concerned to include some test questions when the students write their class test on electricity. The class test was written to test whether the students can transfer what they did in the laboratory to class and whether they can relate theory and practical work.

Interviews. Interviews were selected as one of the research tools in this study. Follow up questions were also taken into consideration when interviews were conducted since "interviews can encourage respondents to develop their own ideas, feelings, insights, expectations or attitudes and it allows respondents to say what they think" (Richardson, n.d.).

Reliability and validity of the instruments

Reliability and validity are crucial criteria in assessing the quality of a research study (Seale, 2004). All the instruments used in a study should be tested for their reliability and their validity as Engelhardt and Beichner (2004, p. 102) stated – for the test to be useful and its results to be accurately applied and interpreted, it must be reliable and valid. Even though the reliability and the validity of the instruments used in this study were tested by their developers, it was taken into consideration that the tests were used in another context, and then reliability and validity were checked. All the tests were given to two physics lecturers and a chemistry lecturer to check the content. The tests were also given to the physics I group to check whether they are familiar with the language used and whether they can finish writing it within the given time frame. Reliability relates to the – extent to which an instrument provides similar results every time it is administered to the same sample at different times (Engelhardt and Beichner, 2004, p. 102). Validity is the – extent to which a test measures what it claims to measure and is not a quality that can be established in a single measurement, but accumulated via several measurements (Engelhardt and Beichner, 2004, p. 103).

Method

Data were collected from two groups. The students were initially given the DIRECT test to evaluate their prior knowledge of electric circuits as well as their understanding. This test was used as baseline information to evaluate their understanding before the electricity module was presented in class. On the test paper, the space was provided where students had to indicate their reason for selecting their answer. Those reasons were analyzed. The second test was TISP to test their integrated science process skills namely; identifying and controlling variables, stating hypotheses, operational definitions, graphing and interpreting data and experimental design. This test does not concentrate on physics only; it concentrates on science in general. The intention was to test whether the students can for example identify the variable given a statement before they do the practical work. Prior to the laboratory sessions, the experimental group was sent to the computer laboratory to do simulations.

Afterwards, both groups did the experiments in the laboratory and completed the same worksheets. While doing the experiments in the laboratory the researcher was observing the two groups by using an observation schedule. DIRECT and TISP were administered as post tests after attending theory classes and both groups completed the practical work. Again their reasons on the DIRECT test were evaluated. A class test which consisted of conceptual questions was written in class. Finally, six students were identified for the interviews, to verify the data collected and were selected based on their performance on the DIRECT pre and post tests.

Results

DIRECT test

The test was a multiple choice consisting of 29 items. Figure 1 indicates the performance of students from the experimental group whereas Figure 2 indicates the performance of the students from the control group for both pre and post tests.

For items 17, 19, 21, 23, and 25, students did better in the pre than in the post-test. The reason might be that when they were writing the post-test, they were also committed to other tests. Another reason might be that the students did not understand the concepts; their reasoning showed misunderstanding especially in items 17 and 23 where they scored zero (0 per cent) in the post-test as compared to the pre test where they scored 10 per cent. The gap between the pre test results and the post-test results for items 1, 2, 3, 9, 10, 11, 13, 18, 24, 26 and 28 is 20 per cent and above. This could indicate an improvement because the students did the simulations, laboratory work as well as attended lectures on electric circuits. Owing to the fact that the experimental group had extra practice on electric circuits, it might have contributed to their improvement.

There were items whereby the students did better in the pre than in the post-test, for example for items 4, 18, 22 and 23. In these cases, the students did not score anything (0 per cent) for the post-test whereas for the pre-test they scored 10-20 per cent. It might be because sometimes students do not read questions for understanding and maybe during the pre-test they did not know the correct answer they just guessed because it was multiple choice questions. In some cases, they did not give an explanation. There was consistency in item 7, because the score was 100 per cent in both the tests as well

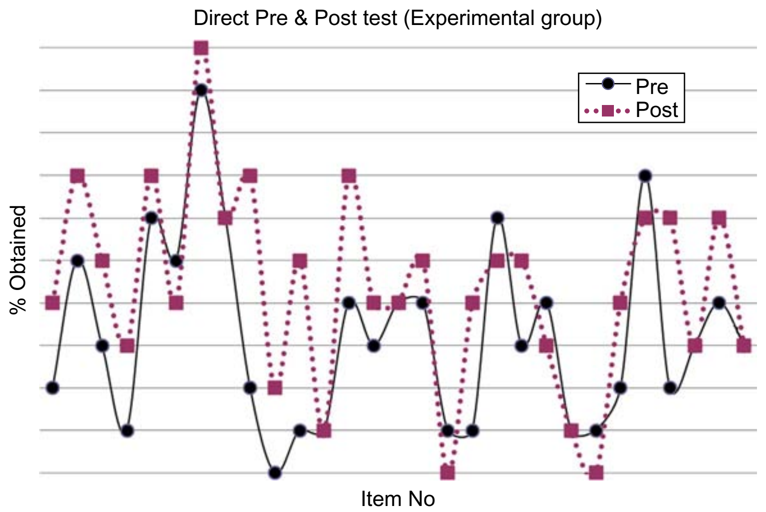


Figure 1.
The performance of experimental group students DIRECT pre and post tests

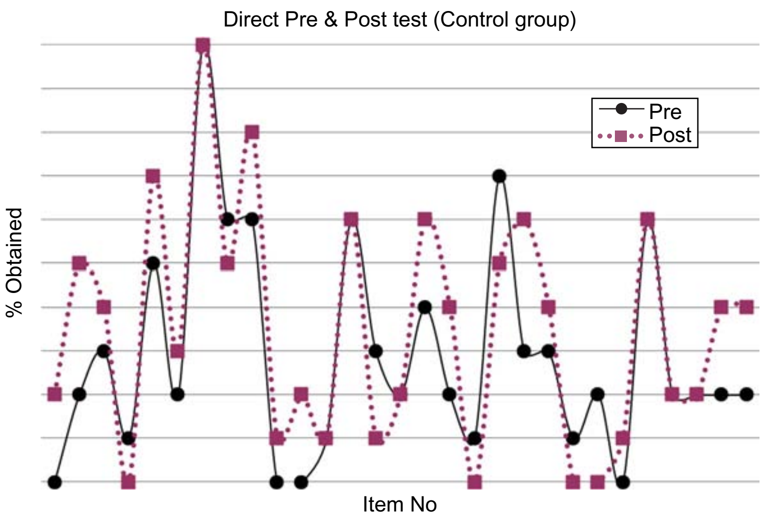


Figure 2.
The performance of control group students on the DIRECT pre and post tests

as for items 13 and 25 where the score was 60 per cent in both tests. The consistency might mean that the students do have an idea of how electric circuit work, they did understand and they have learned.

Item 7 was regarded as being very easy. This was the item whereby the students had to compare the brightness of the bulb in circuit 1 with that in circuit 2 as shown in Figure 3. They had to indicate which bulb was brighter.

All students from the experimental group and 90 per cent from the control group managed to give the correct answer. The reasoning of students was based on the number of batteries that the circuit have: The more the batteries, the more power the circuit will have and therefore it will be more brighter when it has more than one battery.

Therefore, the bulb in circuit 1 is brighter because of two cells. Their reasoning concur with Bryan and Stuessy (2006) in their paper – the Brightness Rules' alternative conception for light bulb circuits, they confirmed that students reasoned that the total brightness available to the bulb or bulbs in a circuit is dependent on the power supply.

Item 22 was a schematic diagram and students had to show the realistic circuit that represents that schematic diagram (Figure 4).

Both groups scored 10 per cent in average. In total, 40 per cent of the experimental group chose circuits C and D. Those who chose circuits C and D said it is because all four bulbs in those circuits are connected in series with the battery, therefore, they represent the schematic diagram. They did not notice the difference between the two circuits. The two bulbs facing down in circuit D are not properly connected. The wire connecting the bulbs should connect at the bottom of the bulb like in circuit C. The reason why students did badly in other items for the post-test was that simulations did not adress all the misconceptions.

TISP pre and post tests

The TISP test was assessing five process skills. These skills are shown in Figures 5 and 6. The experimental group did better than the control group in two skills namely; identifying variables and stating hypothesis. In the other three skills, the control group performed better. During the observations, it was observed that the control group did better than the experimental group in stating the hypothesis whereas the experimental group did better in graphing and interpreting data. The two groups performed the same on the experimental design. In identifying variables and operational definitions, the observation schedule agrees with the TISP results.

Students should develop all science process skills because competence in process skills enables students to learn with understanding. The role of science

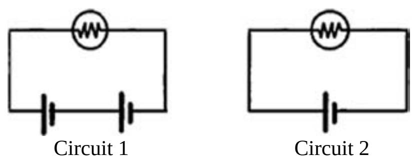


Figure 3.
Circuit diagram

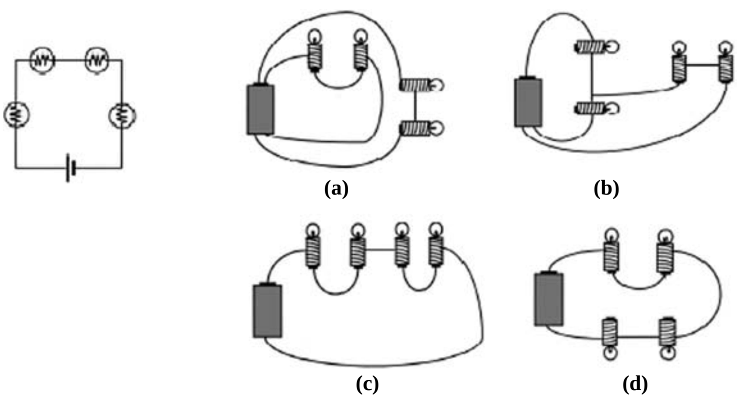


Figure 4.
Schematic diagram of a circuit

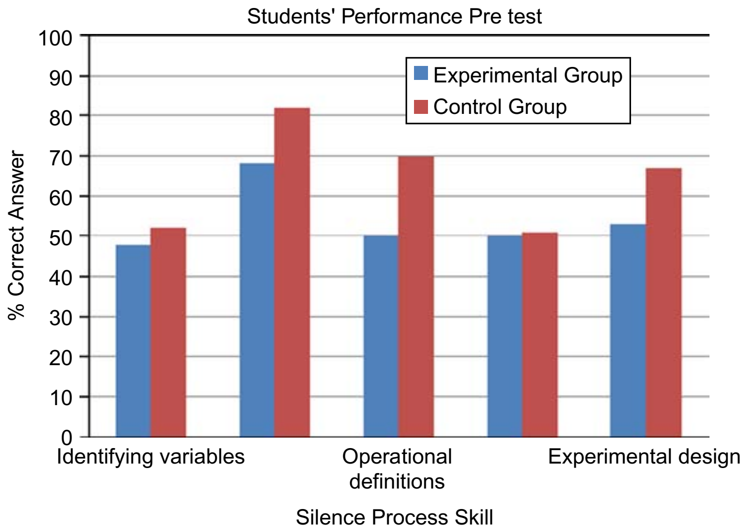


Figure 5.
Science process skills tested during pre-test

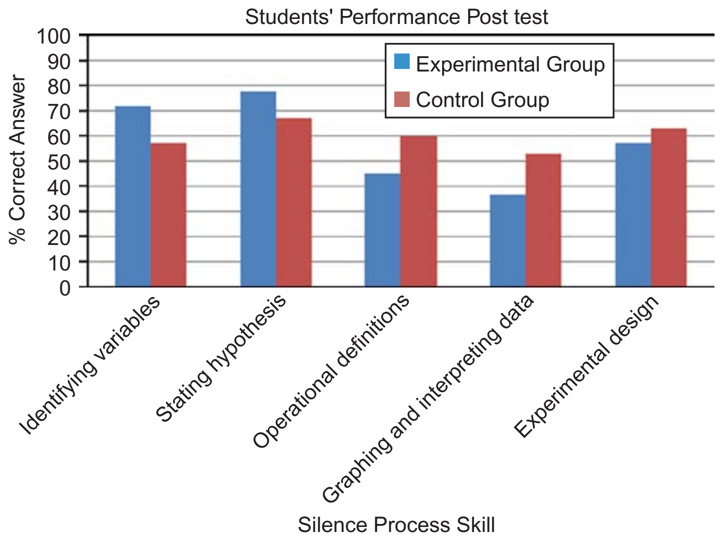


Figure 6.
Science process skills tested during post-test

process skills in the development of learning with understanding is of crucial importance (Kazeni, 2005).

Observations and worksheets

Students were required to do hands on experiments in the laboratory on series and parallel connection of resistors. Apart from doing the experiments, an additional aim of the practical work was to develop the following skills; identifying and controlling variables, experimental design, graphing and interpreting graphs. During the first ten minutes of the session, a demonstration on the expected outcome had to be performed and, thereafter, the students had to perform experiment themselves.

During the practical work, the researcher made some observations on how the students handle the apparatus, how they connect the equipment and how they record data. The main focus was on the five integrated process skills. The two groups were observed and the observation sheet was filled in for each student and the average performance was indicated in Tables I and II. The process skills were rated on a scale of 1 to 5, where 1 is very poor and 5 is very good. The researcher took into consideration that in qualitative research bias affects the validity and reliability of findings; that is the reason why two demonstrators were requested to fill in the observation sheet and the average was determined based on the researcher and demonstrators' sheets. The results are as shown in Tables I and II.

It is shown in both tables that the two groups showed less difference in terms of the development of the process skills. Comparing the observation and the performance on the TISP test, it was shown that the control group performed better than the experimental group. The two groups were required to draw graphs after performing an experiment. They were supposed to draw a graph of current versus the potential difference of the three resistors on the same set of axes and determine the value of each resistor from the graph. Figures 7 and 8 show the example of the graphs drawn by the students from both groups. The two graphs showed a difference in terms of labeling the axes and drawing the straight line.

There were some differences in the way students have drawn the graphs. Others were able to draw the graph and they labelled the graphs accordingly whereas one has drawn three graphs on the same axes but did not label them. It was difficult to see which one is R_1 or R_2. Student in Figure 6 was unable to label the axis correctly. Current is the dependent variable and was supposed to be plotted on the *y* axis and the potential difference as the independent variable was supposed to be plotted on the *x* axis. The problem, when the axis are not labelled correctly, is that they are not going

Table I. Ratings on the observation schedule for experimental group

	Experimental group Process skill	1	2	3	4	5
1	Identifying variables					X
2	Stating hypothesis			X		
3	Operational definitions			X		
4	Graphing and interpreting data				X	
5	Experimental design					X

Table II. Ratings on the observation schedule for control group

	Control group Process skill	1	2	3	4	5
1	Identifying variables				X	
2	Stating hypothesis				X	
3	Operational definitions				X	
4	Graphing and interpreting data			X		
5	Experimental design					X

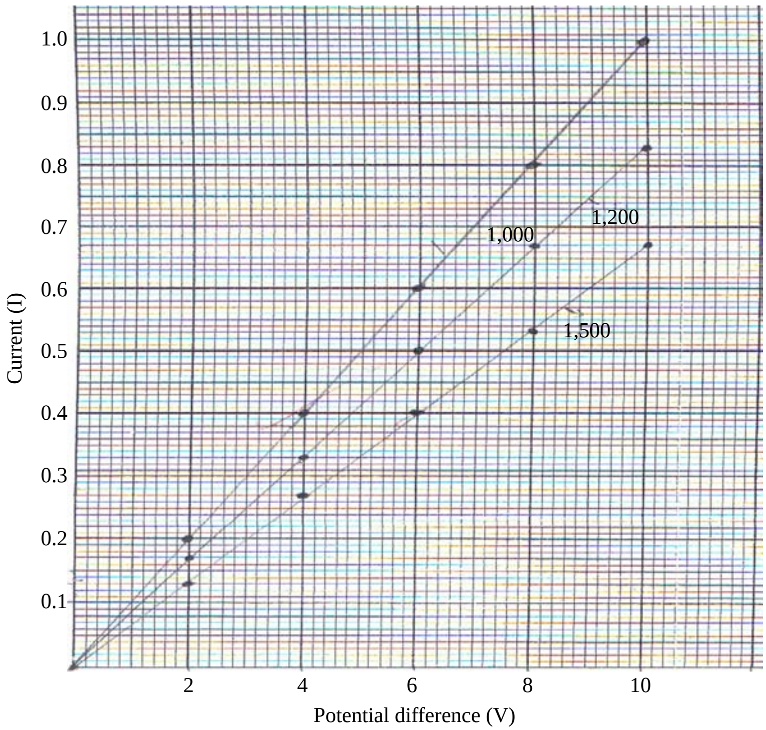


Figure 7.
Example of the graph (Student 1)

to get the correct value of the resistance of the resistor from the graph. The points from the graph were not joined correctly, the student was tracing every point, when was supposed to draw a straight line between the majority of points.

Class test

Students were given a class test after completing the electricity and magnetism module. The reason for giving them a class test was to verify whether they can apply the knowledge they obtained from the laboratory work. In the laboratory students would connect a circuit and if the current is not flowing sometimes they could not identify the problem. The questions were more on conceptual understanding than on calculations.

Figures 9 and 10 show how the students responded to question 1 from the class test. These responses were selected because one student agreed to the question and the other student disagreed. They were chosen to show the difference in interpreting the diagrams and whether the students can give a correct explanation of why they agreed or disagreed.

The summary of the class test is given in Table III. It gives the number of students who got the correct and the wrong answers to all the questions in the class test. The correct response is italicized.

Interviews

Six questions were asked during the interview to the six students, three from each group. The selection of these students was done by considering their scores

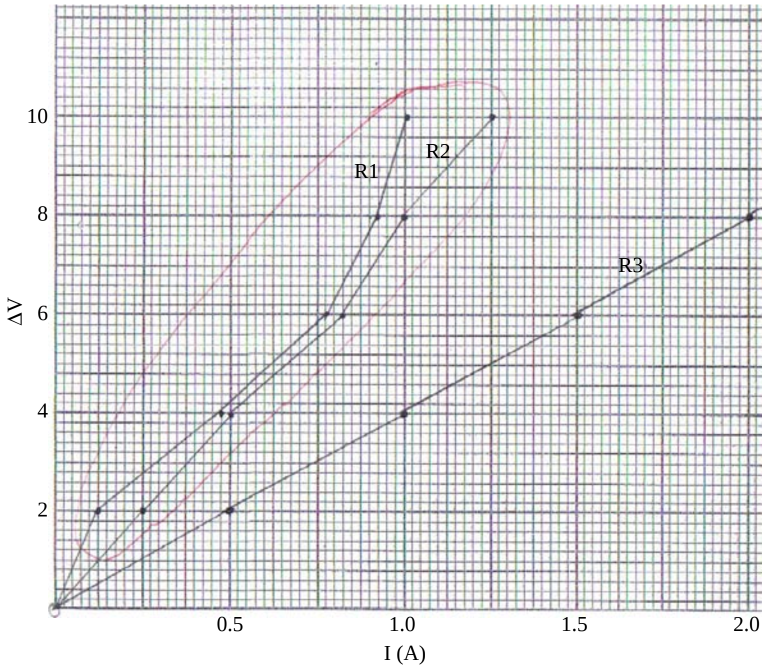


Figure 8.
Example of the graph (Student 2)

Question 1: Is the bulb lit? Yes No Why?

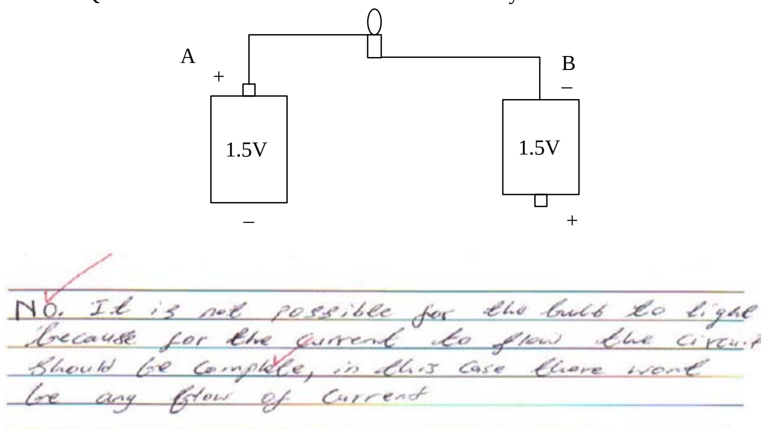
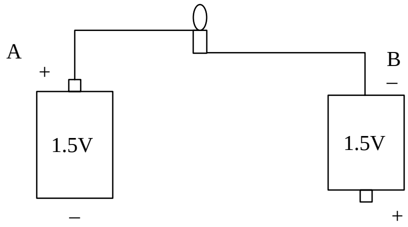


Figure 9.
Example of student's response on the class test

in the pre- and post-test of DIRECT tests. The reason for choosing DIRECT test is because of the conceptual questions. From the six students who were interviewed, one student was able to respond satisfactorily to all the questions and two responded well to five questions out of six. The other three students responded well to the three questions and not satisfactory to the other three. On average, the three students from

Question 1: Is the bulb lit? Yes No Why?



Yes. For the bulb to light it should be connected to a positive terminal and a negative terminal of the battery. It will be very bright because it is connected to two batteries

Figure 10. Example of student's response on the class test

Table III. Performance of students on the class test

	Class test			
	Experimental group		Control group	
Question no.	Yes	No	Yes	No
1	4	*6*	6	*4*
2	*7*	3	*5*	5
3	2	*8*	4	*6*
4	10		10	
5	3	*7*	7	*3*
6	2	*8*	1	*9*

the experimental group were able to respond better than those from the control group. The correct responses (where students were rated good) from the experimental group were 14 out of 18 as compared to 11 for the control group. The ratings of the interviewees are shown in Table IV. The students were rated good (G) if their response were correct and bad (B) if their response were not correct.

Discussions

This study, which investigated the contribution of simulations to the practical work of 20 Foundation Physics students at the University of Limpopo (Medunsa),

Table IV. Rating of six students from the interviews

	Q1		Q2		Q3		Q4		Q5		Q6	
	G	B	G	B	G	B	G	B	G	B	G	B
E_i	X		X		X		X		X		X	
E_s	X		X			X	X		X		X	
E_d		X	X			X	X			X	X	
C_i	X		X			X	X		X		X	
C_s		X	X			X	X			X	X	
C_d		X	X			X	X			X	X	

has not been carried out before. Although the results of this study cannot be generalised due to small sample used, it was the first time that simulations were introduced in the laboratory at Medunsa campus. Simulations can enhance academic performance and learning achievement levels of students (Michael, 2001). The present study reported that the simulations increased the conceptual understanding of students.

The findings of this study confirmed the study conducted by Betz (1996), namely; that the students who supplemented their classes with the use of simulations performed better in their examinations.

After analysis of the data on the 5 process skills collected by TISP pre and post tests, the findings of the study showed that the students from the control group performed better than the students from the experimental group even though the difference is not actually significant (only 1 per cent difference). The results showed that the experimental group did not perform better than the control group even after the simulations. This could indicate that the simulations did not have an effect on the development of science process skills. The observations confirmed the data collected by the TISP tests on the process skills tested. It showed that both groups were doing equally on the process skills; one group was doing well in one skill and the other group on another skill. During the laboratory work the two groups did not show any difference when handling the apparatus and recording the data.

Geban *et al.* (1992) found the impact of simulations on process skills such as, identifying variables, measuring, graphing and interpreting data and designing experiments, to be equally or more valuable than traditional methods. This is in agreement with the findings of this study, because simulations contributed equally to laboratory work with regards to the development of process skills. The simulations did not result in any change on the development of the process skills tested.

Data collected from the interviews was used to supplement the data collected by other methods. The results from the interviews confirmed the results from DIRECT and a class test. The students from the experimental group were able to give valid explanations for example in Figure 9, indicating an understanding of the electric circuits' concepts. Even though simulations increased conceptual understanding, they cannot be a substitute for the work done in the laboratory with real equipment; they should be used jointly because they did not contribute positively to the development of the tested process skills. Higher skills or integrated process skills may be developed using other types of practical work such as laboratory experiments, directed inquiry/problem solving or open inquiry/problem solving. The skill to be developed depends on which experiment is done (Bradley, 2005).

Conclusions

The present study, which investigated the contribution of simulations to the practical work of Foundation Physics students at the University of Limpopo, has not been carried out before. It was the first time that simulations were introduced in the laboratory at Medunsa campus. The present study reported that the simulations increased the conceptual understanding of students, and it agrees with McKee *et al.* (2007). The experimental group showed that the simulations had a constructive contribution on the understanding of electric circuits of students. However; even though simulations increased conceptual understanding, they cannot be a substitute for the work done in the laboratory with real equipment; they should be used jointly because they did not contribute convincingly to the development of the tested process skills.

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